Fair and Load-Balanced Student-Alumni Matching at VJTI

Multi-Objective Capacitated Bipartite Assignment with Rawlsian Fairness & Departmental Parity

Course: Optimization Techniques (University Group Project)
Institution: Veermata Jijabai Technological Institute (VJTI), Mumbai
Notice: ARTIFICIAL - FOR TESTING ONLY (All benchmark data utilize synthetic pseudonymous IDs: `S001` – `S1000`, `A001` – `A500`)

Slide 1: The Problem & Institutional Motivation

The Challenge at VJTI Mumbai:

- Each year, hundreds of graduating engineering students across **8 academic departments** (`CE`, `IT`, `EXTC`, `EE`, `ME`, `CEV`, `PE`, `TT`) seek career mentorship from practicing alumni across 6 industrial domains.
- **Traditional matching** relies on ad-hoc first-come-first-served or greedy score maximization.

Critical Failures of Greedy / Manual Approaches:

1. **The "Superstar Alumnus" Bottleneck**: Top alumni receive overwhelming request loads beyond sustainable capacity, leading to mentor burnout and disengagement.
2. **Peripheral Student Neglect**: Students with average GPAs or non-computing backgrounds receive low-quality matches ($Q_i < \theta$), violating egalitarian welfare.
3. **Cross-Departmental Inequity**: Computing departments (`CE`, `IT`) dominate high-scoring alumni; core engineering departments (`PE`, `CEV`, `TT`, `ME`, `EE`) experience structural deficits due to smaller alumni cohorts.

Slide 2: Mathematical Formulation (MO-CBAP)

Decision Variables:

- $x_{ij} \in {0, 1}$: Binary assignment of student $i \in \mathcal{S}$ to alumnus mentor $j \in \mathcal{A}$.
- $y_j \in {0, 1}$: Binary activation status of alumnus mentor j ($y_j = 1 \iff \sum_i x_{ij} \ge 1$).

The Tri-Objective Vector:

$$\min_{\mathbf{x}, \mathbf{y}} \mathbf{F}(\mathbf{x}) = \begin{bmatrix} -Z_1(\mathbf{x}) \ -Z_2(\mathbf{x}) \ +Z_3(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} -\sum_{i=1}^N \sum_{j=1}^M q_{ij} x_{ij} \ -\min_{i \in \mathcal{S}} Q_i \ +\max_{j \in \mathcal{A}} L_j \end{bmatrix}$$

1. **Utilitarian Efficiency (Z_1)**: Maximize total cumulative match score across all students.
2. **Rawlsian Egalitarian Fairness (Z_2)**: Maximize the match quality of the least-advantaged student ($\min_i Q_i$).
3. **Workload Balancing (Z_3)**: Minimize the peak mentorship load across active alumni mentors ($\max_j L_j$).

Slide 3: Operational Constraints

Constraint	Mathematical Formulation	Operational Purpose at VJTI
Exact Degree (K)	$\sum_j x_{ij} = K, \quad \forall i$	Exactly $K=2$ distinct mentors per student
Alumni Capacity (C_j)	$\sum_i x_{ij} \le C_j y_j, \quad \forall j$	Mentor student load cannot exceed declared cap
Consent (c_j)	$x_{ij} \le c_j, \quad \forall i, j$	Mandatory opt-in prior consent ($c_j \in {0, 1}$)
Quality Threshold ($q_{\min}$)	$x_{ij} = 0 \text{ if } q_{ij} < q_{\min}$	Hard cutoff preventing poor-quality pairings ($q_{\min} = 0.25$)
Student Quality Floor (θ)	$Q_i = \sum_j q_{ij} x_{ij} \ge \theta, \quad \forall i$	Guarantees minimum aggregate mentorship quality ($\theta = 0.70$)
Branch Parity (δ)	$	\overline{Q}*b* - *overline{Q}*{b'}
Corporate Diversity Cap	$\sum_{j \in \mathcal{A}*k* x*{ij} \le 1, \quad \forall i, k$	Prevents student clustering under a single employer firm

Slide 4: Algorithmic Architecture

Our methodology benchmarks three methodological tiers:



Slide 5: The Two-Pass `MatchingRepair` Mechanism

Why Generic Evolutionary Operators Fail:

- Discrete integer chromosomes $\mathbf{X} \in [0, M-1]^{N \times K}$ produce duplicate alumni, capacity overflows, and diversity violations during standard crossover and mutation.

Two-Pass Repair Mechanism:

- **Pass 1: Student-Level Feasibility**:
- Removes duplicate assignments within student row i.
- Enforces consent ($c_j = 1$) and match score threshold ($q_{ij} \ge q_{\min}$).
- Resolves employer diversity violations (at most 1 mentor per company).
- **Pass 2: System-Level Workload Re-balancing**:
- Detects overloaded alumni ($L_j > C_j$).
- Systematically swaps students from overloaded mentors to underutilized candidates with the highest available q_{ij}, preserving quality floors and branch parity bounds.

Slide 6: Small Instance Benchmark (30 Seeds, Shared Reference HV)

Rigorous multi-seed evaluation on `small_50x30` across 30 independent runs under a single shared reference point $\mathbf{r}_{\text{shared}} = [-36.0, -0.45, 8.80]$:

Method	Hypervolume Mean $\pm$ Std	Total Quality Z_1	Worst Student Z_2	Max Load Z_3	Spacing	Feasible Rate	Runtime (s)
Random	0.0117 $\pm$ 0.0084	44.79	0.55	7.07	0.000	0.0%	0.15s
Greedy	0.0932 $\pm$ 0.0000	72.62	1.02	8.00	0.000	100.0%	**0.003s**
Min-Cost Flow (raw)	0.0952 $\pm$ 0.0000	73.35	1.02	8.00	0.000	0.0% ($CV=6.0$)	0.05s
MCF Repaired	0.0929 $\pm$ 0.0000	72.51	1.02	8.00	0.000	100.0%	0.05s
NSGA-II	0.3648 $\pm$ 0.0313	65.11	0.94	6.00	0.381	100.0%	17.89s
SPEA2	0.3739 $\pm$ 0.0384	65.18	0.95	5.97	0.481	100.0%	17.96s
MOEA/D	0.2242 $\pm$ 0.0552	63.02	0.78	6.10	0.008	89.4%	26.32s
Hybrid Memetic	**0.7650 $\pm$ 0.1335**	**72.62**	**1.13**	**5.53**	**0.622**	**75.9%**	**31.96s**

- **Wilcoxon Signed-Rank Tests**: Hybrid strictly dominates NSGA-II, SPEA2, and MOEA/D ($p = 1.8626 \times 10^{-9}$, rank-biserial $r = 1.0$).

Slide 7: Exact HiGHS MILP Ground Truth & Secondary Scenario

Exact Theoretical Ground Truth (`small_50x30`):

- Exact solver identified **15 mathematically proven non-dominated Pareto points**:
- Quality Extremum: $Z_1 = 73.1377, \; Z_2 = 1.0230, \; Z_3 = 8$ (26 active alumni, $H = 3.054$).
- Balanced Mid-Point: $Z_1 = 71.0318, \; Z_2 = 1.0125, \; Z_3 = 4$ (27 active alumni, $H = 3.127$).
- Rawlsian Extremum: $Z_1 = 62.9156, \; Z_2 = 1.1828, \; Z_3 = 4$ (28 active alumni, $H = 3.226$).

Secondary Scenario: Active Alumni Minimization ($\min \sum y_j$):

- **At $\ge 80\%$ Max Quality**: Only **15 alumni** needed ($Z_1 = 59.04, Z_2 = 0.75, Z_3 = 8, H = 2.75$).
- **At $\ge 90\%$ Max Quality**: Still only **15 alumni** needed ($Z_1 = 66.02, Z_2 = 0.72, Z_3 = 8, H = 2.60$).
- **At $\ge 95\%$ Max Quality**: **16 alumni** required ($Z_1 = 69.53, Z_2 = 1.02, Z_3 = 8, H = 2.85$).
- **Administrative Value**: Engaging just **50% of alumni roster** captures **90% of total possible mentorship value**.

Slide 8: Scalability Benchmark Across Dimensions

Benchmarking across Small (50×30), Medium (300×150), and Large (1000×500):

Scale	Demand ($N \times K$)	Greedy Runtime	Min-Cost Flow	NSGA-II Runtime	Hybrid Memetic Runtime
Small (50×30)	100 matches	0.007s	0.133s	4.01s	4.34s
Medium (300×150)	600 matches	0.126s	0.969s	26.11s	19.79s
Large (1000×500)	2000 matches	1.248s	13.032s	61.71s	92.53s

Performance on Large Instance (1000×500):

- **Greedy Baseline**: $Z_1 = 1792.91, \; Z_2 = 1.32, \; Z_3 = 10, \; \text{Gini} = 0.0580$ (0.74s).
- **NSGA-II**: $HV = 845.95, \; Z_1 = 1624.10, \; Z_2 = 1.05, \; \text{Gini} = 0.1524$ (20.01s).
- **Hybrid Memetic NSGA-II**: **$HV = 2297.35$ (+171.6%)**, **$Z_1 = 1801.40$**, **$Z_2 = 1.38$**, **$\text{Gini} = 0.0435$ (3.5x lower inequality)**.

Slide 9: Algorithmic Component Ablation Study

Isolating the contributions of heuristic seeding and Lamarckian local search over 30 independent seeds:

Architecture Variant	Hypervolume Mean $\pm$ Std	Total Quality Z_1	Worst Student Z_2	Max Load Z_3	Spacing	Feasible Rate	Runtime (s)
Seeding Only	0.6307 $\pm$ 0.0641	72.62	1.10	6.00	0.685	68.39%	16.98s
Local Search Only	0.3759 $\pm$ 0.1075	66.88	1.07	5.07	0.447	**100.0%**	26.97s
Full Hybrid (Both)	**0.8704 $\pm$ 0.1577**	**72.64**	**1.11**	**5.47**	**0.652**	**77.21%**	**35.33s**

Statistical Significance:

- Full Hybrid vs. Seeding Only: $p = 9.3132 \times 10^{-9}, \; r = 0.9871$ (Statistically significant).
- Full Hybrid vs. Local Search Only: $p = 1.8626 \times 10^{-9}, \; r = 1.0000$ (Statistically significant).
- **Finding**: Seeding provides the critical initial quality boost ($Z_1: 66.88 \to 72.62$), while local search broadens Pareto spread and suppresses peak workload.

Slide 10: Parameter Sensitivity & The Price of Parity (δ)

How much utilitarian efficiency does VJTI sacrifice to guarantee equal departmental outcomes?

Department Disparity Tolerance δ	Max Achievable Quality Z_1	Achieved Disparity	Efficiency Loss vs Baseline	Feasibility Rate
$\delta = 0.15$ (Ultra-Strict Parity)	72.62	0.148	-9.0% quality penalty	0.0% (Infeasible)
$\delta = 0.25$ (Balanced Parity)	72.62	0.244	-3.9% quality penalty	75.0%
$\delta = 0.35$ (Standard Parity)	72.62	0.326	0.0% (Reference)	80.0%
$\delta = 0.50$ (Relaxed Parity)	72.62	0.428	-2.4% quality penalty	75.0%

- **Institutional Recommendation**: Setting $\delta = 0.25$ provides an optimal compromise, eliminating perceived bias against core engineering departments at an efficiency cost of only 3.9%.
- [TODO: Human Interpretation Required — Institutional confirmation of $\delta = 0.25$ by VJTI Departmental Heads].

Slide 11: Institutional Coverage Gaps & Supply-Demand Deficits

Analysis of departmental mentor supply vs student demand on `small_50x30` (`results/coverage_gap.csv`):

Department / Category	Student Demand ($N \cdot K$)	Alumni Supply Capacity	Unmet Demand Deficit	Supply/Demand Ratio	Deficit Severity Score
Civil Engineering (`CEV`)	20 matches (10 students)	7 slots (1 alumnus)	**13**	**0.350**	**14.32 (Critical)**
Production Engineering (`PE`)	24 matches (12 students)	18 slots (4 alumni)	**6**	**0.750**	**6.00 (High)**
Electrical Engineering (`EE`)	26 matches (13 students)	23 slots (4 alumni)	**3**	**0.885**	**2.82 (Moderate)**
Consulting & Strategy Domain	16 matches (8 students)	7 slots (2 alumni)	**9**	**0.438**	**11.06 (Critical)**

- **Actionable Insight**: The placement office must conduct targeted alumni outreach in Civil and Production Engineering prior to allocation rounds.
- [TODO: Human Review Required — Validate alumni recruitment targets with VJTI Alumni Association].

Slide 12: Pareto Frontier Interpretation & Operating Policies

From `results/interpretation_small_50x30.md`: Multi-objective optimization identifies four distinct operational policies on `small_50x30`:

Policy Profile	Total Quality Z_1	Worst Student Z_2	Max Load Z_3	Mean Quality $\overline{Q}$	Gini Index	Active Alumni	Entropy H	Feasible?
1. Highest Quality	**72.62**	1.08	7	1.45 $\pm$ 0.19	0.0741	28 (93.3%)	3.001	No
2. Fairest Policy	71.63	**1.09**	6	1.43 $\pm$ 0.21	0.0823	28 (93.3%)	3.067	**Yes**
3. Lightest Load	68.31	1.08	**5**	1.37 $\pm$ 0.19	0.0800	28 (93.3%)	**3.213**	**Yes**
4. Knee Point (Recommended)	**71.63**	**1.09**	**6**	**1.43 $\pm$ 0.21**	**0.0823**	**28 (93.3%)**	**3.067**	**Yes**

- **Strategic Value of Knee Point**: Preserves **98.6% of peak quality** ($Z_1 = 71.63$) while guaranteeing **100% of peak Rawlsian fairness** ($Z_2 = 1.09$) and capping peak load at 6 students across 28 active mentors.
- [TODO: Human Interpretation Required — Formal administrative sign-off on Knee Point policy].

Slide 13: Honest Assessment of Methodological Limitations

1. **Synthetic Data**:
   - All student/alumni records and compatibility matrices are generated via parametric models.
   - Synthetic data cannot account for interpersonal chemistry, soft skill dynamics, or qualitative student preferences.
2. **NSGA-II vs. Greedy Priority Baseline**:
   - Unseeded NSGA-II attains lower total quality ($Z_1 = 65.11$) than Greedy ($Z_1 = 72.62$) and requires ~18s vs. 0.003s.
   - Unseeded evolutionary operators struggle in constrained discrete assignment spaces without heuristic seeding.
3. **Min-Cost Network Flow Infeasibility**:
   - Raw MCF achieves peak quality ($Z_1 = 73.35$) but is inherently infeasible ($CV = 6.00$) due to inability to enforce employer diversity caps.
4. **Single-Run Exact Solver Limits**:
   - HiGHS MILP provides guaranteed Pareto ground truth, but is evaluated as a single deterministic run on `small_50x30` only (14.25s); it cannot scale to medium or large cohorts.
5. **[TODO: Human Review Required — Validate qualitative mentor-mentee feedback mechanisms]**.

Slide 14: Institutional Recommendations & Implementation Roadmap

Concrete Action Steps for VJTI Placement & Mentorship Office:

1. **Deploy Seeded Memetic NSGA-II**: Integrate solver into the VJTI Alumni Portal backend to generate provably feasible multi-objective matchings in under 45 seconds for 1,000 students.
2. **Standardize Departmental Parity at $\delta = 0.25$**: Eliminates cross-departmental inequity for core engineering branches (`CEV` , `PE` , `TT`) while incurring less than 4% quality loss.
3. **Institutionalize Corporate Diversity Cap ($\text{cap}_{\text{comp}} = 1$)**: Prevents student clustering under single employer firms, ensuring multifaceted career mentorship.
4. **Execute Targeted Alumni Outreach**: Mobilize alumni associations to recruit mentors in Civil Engineering (deficit = 13) and Production Engineering (deficit = 6) before matching rounds.
5. **[TODO: Human Decision Required — Final approval of allocation policy by Dean of Academic Programs]**.